MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 11** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM:** 30

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Periodic Test 11

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. If $u_i = \dfrac{x_i - 25}{10}$, $\sum f_i u_i = 20$, and $\sum f_i = 100$, then the value of the mean $\bar{x}$ is: [1]

(A) 27 (B) 25 (C) 30 (D) 35

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 27

Justification: Comparing $u_i = \frac{x_i - a}{h}$, we have assumed mean $a = 25$ and class size $h = 10$.

$$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) \times h = 25 + \left(\frac{20}{100}\right) \times 10 = 25 + 2 = \mathbf{27}.$$

$$\boxed{\bar{x} = \mathbf{27}}$$

Q2. For the following data: [1]

Marks	0 – 10	10 – 20	20 – 30	30 – 40
Students	4	9	15	12

The modal class and median class respectively are:

(A) 20 – 30 and 20 – 30 (B) 10 – 20 and 20 – 30
(B) 20 – 30 and 30 – 40 (D) 30 – 40 and 20 – 30

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 20 – 30 and 20 – 30

Justification: Highest frequency is 15 $\implies$ Modal Class = **20 – 30**.

Total frequency $N = 4 + 9 + 15 + 12 = 40 \implies \dfrac{N}{2} = 20$.

Cumulative frequencies: 0 – 10 : 4; 10 – 20 : 13; 20 – 30 : 28; 30 – 40 : 40.

The cumulative frequency just greater than 20 is 28 $\implies$ Median Class = **20 – 30**.

Q3. If $x_1, x_2, \ldots, x_n$ are n observations with mean $\bar{x}$, then the mean of $ax_1 + b, ax_2 + b, \ldots, ax_n + b$ is: [1]

(A) $a\bar{x}$ (B) $\bar{x} + b$ (C) $a\bar{x} + b$ (D) $\frac{\bar{x}+b}{a}$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (C) $a\bar{x} + b$

Justification:

$$\bar{y} = \frac{\sum(ax_i + b)}{n} = \frac{a\sum x_i + nb}{n} = a\left(\frac{\sum x_i}{n}\right) + b = \mathbf{a\bar{x} + b}.$$

$$\boxed{\text{New Mean} = \mathbf{a\bar{x} + b}}$$

Q4. If the mode of a series exceeds its mean by 12, then the mode exceeds the median by: [1]

(A) 4 (B) 8 (C) 6 (D) 10

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 8

Justification: From empirical relationship:

$$\text{Mode} - \text{Mean} = 3(\text{Median} - \text{Mean})$$

$$\text{Given Mode} - \text{Mean} = 12 \implies 12 = 3(\text{Median} - \text{Mean}) \implies \text{Median} - \text{Mean} = 4.$$

$$\text{Mode} - \text{Median} = (\text{Mode} - \text{Mean}) - (\text{Median} - \text{Mean}) = 12 - 4 = \mathbf{8}.$$

$$\boxed{\text{Mode} - \text{Median} = \mathbf{8}}$$

Q5. The class mark of the class $90 - 120$ is: [1]

(A) 90 (B) 105 (C) 115 (D) 120

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 105

Justification:

$$\text{Class Mark} = \frac{\text{Upper Limit} + \text{Lower Limit}}{2} = \frac{120 + 90}{2} = \frac{210}{2} = \mathbf{105}.$$

$$\boxed{\text{Class Mark} = \mathbf{105}}$$

Q6. **Assertion (A):** The mode of a grouped distribution can be calculated directly even when class intervals are discontinuous $(1 - 10, 11 - 20)$.
Reason (R): The lower limit (l) and class width (h) in the mode formula must be determined from continuous true class limits. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (D) (A) is false but (R) is true.

Justification: The formula for mode cannot be applied directly to discontinuous/inclusive class intervals; they must first be converted into continuous class boundaries. Hence, Assertion (A) is false. Reason (R) correctly specifies that l and h must come from continuous true limits, so Reason (R) is true.

SECTION B: Very Short Answer Type

MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. Find the median of all prime numbers less than 20. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Listing Primes and Count [1 Mark]

The prime numbers strictly less than 20 are: $2, 3, 5, 7, 11, 13, 17, 19$.

Total number of observations $n = 8$ (even).

Step 2: Median Calculation [1 Mark]

$$\text{Median} = \frac{\left(\frac{n}{2}\right)\text{ th term} + \left(\frac{n}{2}+1\right)\text{ th term}}{2} = \frac{4\text{th term} + 5\text{th term}}{2} = \frac{7+11}{2} = \frac{18}{2} = \mathbf{9}.$$

$$\boxed{\text{Median} = \mathbf{9}}$$

Q8. The mean of 5 numbers is 18. If one number is excluded, their mean becomes 16. Find the excluded number. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Finding Total Sums [1 Mark]

Sum of all 5 numbers $= 5 \times 18 = 90$.

Sum of remaining 4 numbers after exclusion $= 4 \times 16 = 64$.

Step 2: Excluded Number Calculation [1 Mark]

$$\text{Excluded Number} = 90 - 64 = \mathbf{26}.$$

$$\boxed{\text{Excluded Number} = \mathbf{26}}$$

SECTION C: Short Answer Type

MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. Find the median height of 60 students from the following data: [3]

Height (cm)	135 – 140	140 – 145	145 – 150	150 – 155	155 – 160	160 – 165
Students	4	9	18	16	10	3

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Cumulative Frequency Table [1 Mark]

Height (cm)	Number of Students (f_i)	Cumulative Frequency (cf)
$135-140$	4	4
$140-145$	9	$4+9=13$
$145-150$	18	$13+18=31$
$150-155$	16	$31+16=47$
$155-160$	10	$47+10=57$
$160-165$	3	$57+3=60$
Total	$N=60$	

Step 2: Median Class & Parameters [1 Mark]

$N=60 \implies \dfrac{N}{2}=30$.

The cumulative frequency just greater than 30 is 31, corresponding to class **$145-150$**.

$\therefore$ Median Class $= 145-150$.

Lower limit $(l)=145$, $cf=13$, frequency $(f)=18$, class width $(h)=5$.

Step 3: Median Calculation [1 Mark]

$$\text{Median} = l+\left(\frac{\frac{N}{2}-cf}{f}\right)\times h = 145+\left(\frac{30-13}{18}\right)\times 5 = 145+\frac{85}{18} \approx 145+4.72 = \mathbf{149.72}\text{ cm}.$$

$$\boxed{\text{Median Height} = \mathbf{149.72}\text{ cm} \quad \left(\text{or } \mathbf{149\frac{13}{18}}\text{ cm}\right)}$$

Q10. Find the mean of the following distribution using Assumed Mean method ($a=32.5$): [3]

Class	$15-20$	$20-25$	$25-30$	$30-35$	$35-40$	$40-45$
Frequency	4	6	10	13	5	2

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Assumed Mean Calculation Table [1.5 Marks]

Let Assumed Mean $a=32.5$, $d_i=x_i-32.5$.

Class	Frequency (f_i)	Class Mark (x_i)	$d_i=x_i-32.5$	f_id_i
$15-20$	4	17.5	-15	-60
$20-25$	6	22.5	-10	-60
$25-30$	10	27.5	-5	-50
$30-35$	13	32.5	0	0
$35-40$	5	37.5	$+5$	$+25$
$40-45$	2	42.5	$+10$	$+20$
Total	$\sum f_i=40$			$\sum f_id_i=-125$

Step 2: Formula Declaration & Substitution [1 Mark]

$$\bar{x} = a+\frac{\sum f_id_i}{\sum f_i} = 32.5+\frac{-125}{40} = 32.5-3.125 = \mathbf{29.375} \quad \left(\text{or } 29\frac{3}{8}\right)$$

Step 3: Final Boxed Result [0.5 Mark]

$$\boxed{\text{Mean } (\bar{x}) = \mathbf{29.375} \quad \left(\text{or } \mathbf{29\frac{3}{8}}\right)}$$

SECTION D: Long Answer Type

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Q11. Find the mean and mode of the following distribution of daily pocket allowance of 64 children: [5]

Allowance (Rs)	$11-13$	$13-15$	$15-17$	$17-19$	$19-21$	$21-23$	$23-25$
Children	7	6	9	13	20	5	4

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Part 1: Mean Calculation (Assumed Mean Method, $a = 18$) [2.5 Marks]

Allowance (Rs)	**Children (f_i)**	**Class Mark (x_i)**	$d_i = x_i - 18$	$f_i d_i$
$11-13$	7	12	-6	-42
$13-15$	6	14	-4	-24
$15-17$	9	16	-2	-18
$17-19$	13	18	0	0
$19-21$	20	20	$+2$	$+40$
$21-23$	5	22	$+4$	$+20$
$23-25$	4	24	$+6$	$+24$
Total	$\sum f_i = 64$			$\sum f_i d_i = 0$

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i} = 18 + \frac{0}{64} = \mathbf{18} \text{ Rs.}$$

Part 2: Mode Calculation [2.5 Marks]

The highest frequency is 20, which lies in modal class $\mathbf{19-21}$.
$l = 19, h = 2, f_1 = 20, f_0 = 13, f_2 = 5$.

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h = 19 + \left(\frac{20-13}{2(20)-13-5}\right) \times 2 = 19 + \left(\frac{7}{22}\right) \times 2$$

$$\text{Mode} = 19 + \frac{7}{11} = 19 + 0.636 \approx \mathbf{19.64} \text{ Rs} \quad \left(\text{or } 19\frac{7}{11} \text{ Rs}\right).$$

$$\boxed{\text{Mean} = \text{Rs } \mathbf{18}, \quad \text{Mode} = \text{Rs } \mathbf{19.64} \quad \left(\text{or Rs } \mathbf{19\frac{7}{11}}\right)}$$

Q12. The following distribution gives the state-wise teacher-student ratio in higher secondary schools of India. Find the mode and mean of this data: [5]

Ratio	$15-20$	$20-25$	$25-30$	$30-35$	$35-40$	$40-45$	$45-50$	$50-55$
States	3	8	9	10	3	0	0	2

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► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Part 1: Mode Calculation [2.5 Marks]

The maximum frequency is 10, which lies in modal class **30 – 35**.
$l = 30, h = 5, f_1 = 10, f_0 = 9, f_2 = 3$.

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h = 30 + \left(\frac{10 - 9}{2(10) - 9 - 3}\right) \times 5 = 30 + \frac{5}{8} = \mathbf{30.63} \quad (\text{or } 30.625).$$

Part 2: Mean Calculation (Step-Deviation Method, $a = 32.5, h = 5$) [2.5 Marks]

Ratio	States (f_i)	Class Mark (x_i)	$u_i = \frac{x_i - 32.5}{5}$	$f_i u_i$
15 – 20	3	17.5	−3	−9
20 – 25	8	22.5	−2	−16
25 – 30	9	27.5	−1	−9
30 – 35	10	32.5	0	0
35 – 40	3	37.5	+1	+3
40 – 45	0	42.5	+2	0
45 – 50	0	47.5	+3	0
50 – 55	2	52.5	+4	+8
Total	$\sum f_i = 35$			$\sum f_i u_i = -23$

$$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) \times h = 32.5 + \left(\frac{-23}{35}\right) \times 5 = 32.5 - \frac{23}{7} = 32.5 - 3.286 \approx \mathbf{29.21} \quad \left(\text{or } 29\frac{3}{14}\right).$$

Mode = **30.63**, Mean = **29.21**

SECTION E: Case Study Based Question

MATHS MANTHAN | (1 × 4 = 4 Marks)

Q13. CASE STUDY: RURAL LIVELIHOOD MISSION SELF-HELP GROUP WEEKLY SAVINGS [4 Marks]

A microfinance cooperative audited the weekly deposit savings (in Rupees) of 120 women members of rural self-help groups (SHGs) under a financial inclusion scheme.

Weekly Savings (Rs)	50 – 100	100 – 150	150 – 200	200 – 250	250 – 300
SHG Members	15	25	40	30	10

(i) Calculate the median weekly savings of the cooperative members. [2]

(ii) Identify the modal savings bracket. [1]

(iii) Find the cumulative frequency of members saving less than Rs 200 weekly. [1]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Subpart (i): Median Weekly Savings [2 Marks]

Total members $N = 120 \implies \dfrac{N}{2} = 60$.

Cumulative frequencies: $50 - 100 : 15$; $100 - 150 : 40$; $150 - 200 : 80$; $200 - 250 : 110$; $250 - 300 : 120$.

The cumulative frequency just greater than 60 is 80, corresponding to median class **150 – 200**.

$l = 150, cf = 40, f = 40, h = 50$.

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h = 150 + \left(\frac{60 - 40}{40}\right) \times 50 = 150 + \left(\frac{20}{40}\right) \times 50 = 150 + 25 = \mathbf{175} \text{ Rs.}$$

Median Weekly Savings = Rs **175**

Subpart (ii): Modal Savings Bracket [1 Mark]

The maximum frequency is 40, corresponding to the bracket **150 – 200** Rs.

Modal Savings Bracket = Rs **150 – 200**

Subpart (iii): Cumulative Frequency (< Rs 200) [1 Mark]

Sum of members saving in $[50 - 100] + [100 - 150] + [150 - 200] = 15 + 25 + 40 = \mathbf{80}$.

Cumulative Frequency (< Rs 200) = **80**

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